MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special

Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third Semester Discipline Specific Elective (DSE), MG3DSECHE201–Introduction to Nanoscience

(2024 ADMISSION ONWARDS)

Question Paper code: 25901889

Instruction to the Examiners:
• This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.
• Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
• Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students

Part A

Short Answer Type Questions

Answer any 5 questions. Each question carries 4 marks

1. **How do magnetotactic bacteria utilize magnetic nanoparticles to orient themselves in their environment?**
   Magnetosomes present in magnetotactic bacteria. Magnetosomes contain magnetite (Fe_3O_4) or greigite (Fe_3S_4). Align along Earth's magnetic field like a natural compass. Helps bacteria reach favourable oxygen environment/orientation.
   (Relevant explanation - 4 marks)

2. **What are the different type of signals produced when an electron beam interacts with sample in electron microscopy?**
   Secondary electrons, Backscattered electrons, Characteristic X-rays, Auger electrons / transmitted electrons / cathodoluminescence
   (Any 4 type of signals - 4 marks)

3. **Define magnetic nanoparticles and give two examples.**
   Magnetic nanoparticles are nanoparticles (size **1–100 nm**) made of magnetic materials that exhibit magnetic properties. Example: Fe_3O_4, γ-Fe_2O_3, Co, Ni
   (Definition with any two examples – 4 marks)

4. **What are the benefits of using nanoparticles in medicinal therapy?**
   Targeted drug delivery, controlled release, improved bioavailability, reduced side effects, imaging, cancer therapy
   (Any 4 benifits - 4 marks)

5. **Describe different types of nanomaterials based on their dimensionality.**
   Zero-dimensional (0D): All three dimensions are in the nanoscale (1–100 nm). Examples: nanoparticles, quantum dots, fullerenes, dendrimers.
   One-dimensional (1D): Two dimensions are in the nanoscale and one dimension is >100 nm. Examples: carbon nanotubes, nanowires, nanorods, nanofibres.
   Two-dimensional (2D): One dimension is in the nanoscale and two dimensions are >100 nm. Examples: graphene, graphyne, borophene, phosphorene, nanofilms and nanocoatings.

Three-dimensional (3D): All three dimensions are >100 nm, but the material is composed of nanoscale building blocks or possesses nanoscale features. Examples: nanocomposites, nanoparticle dispersions, bundles of nanotubes/nanowires.

(Relevant explanation - 4 marks)

6. **Explain the role of carbon dots in bioimaging.**

Carbon dots (CDs) are fluorescent carbon nanoparticles of size. They exhibit strong and tunable photoluminescence (fluorescence), making them excellent fluorescent probes for imaging. they are biocompatible, water-soluble and possess low toxicity, making them safe for biological applications.

(Relevant explanation - 4 marks)

7. **How do nanomaterials facilitate data storage?**

Magnetic nanoparticles are used as high-density storage media, allowing more data to be stored in a smaller area. Their small size enables miniaturization of storage devices with increased storage capacity. Nanomaterials provide faster data writing and reading due to their unique magnetic/electronic properties.

(Relevant explanation - 4 marks)

Part B
Short Essay Type Questions
Answer any 5 questions. Each question carries 7 marks

8. **Describe the key characteristics of metal oxide nanoparticles and their significance.**

Nanoparticles of metal oxides (1–100 nm). They have high surface area-to-volume ratio. They have unique optical/electrical/magnetic/catalytic properties. Examples: TiO_2, ZnO, Fe_3O_4/Fe_2O_3 . Applications: Catalysts, photocatalysts, gas sensors, environmental remediation etc. (any one).

(Definition -2 mark, any two characteristics-2 marks, any one example-1 mark, any one Application/significance-2 marks)

9. **How is nanotechnology applied in hydrogen storage materials and devices?**

Nanomaterials such as carbon nanotubes (CNTs), graphene, metal hydrides and metal–organic frameworks (MOFs) provide a high surface area for efficient hydrogen storage by adsorption (physisorption/chemisorption). They improve hydrogen storage capacity and enable faster hydrogen adsorption and desorption, making storage more efficient. Hydrogen storage nanomaterials are widely used in hydrogen fuel cells and other clean energy devices.

(Relevant explanation with example- 7 marks)

10. **How does nanotechnology help to reduce pollutants in air pollution control systems?**

Nanomaterials such as TiO_2 nanoparticles, nanocatalysts and nanoporous adsorbents efficiently remove or degrade gaseous pollutants and volatile organic compounds (VOCs) due to their high surface area and catalytic activity. Nanofilters and nanofibrous membranes effectively trap particulate matter, toxic gases and other airborne contaminants.

(Relevant explanation with example- 7 marks)

11. **Explain the different types of nanomaterials: (a) carbon-based, (b) metal-based, (c) dendrimers, and (d) composites. Comment on their main properties and applications.**

Carbon-based nanomaterials: Nanomaterials composed entirely of carbon atoms; **examples:** carbon nanotubes (CNTs), graphene, fullerenes.

Metal-based nanomaterials: Nanomaterials composed of metals or metal oxides with at least one dimension in the nanoscale; examples: Au, Ag, TiO_2, ZnO nanoparticles.

Dendrimers: Highly branched, tree-like synthetic polymeric nanomaterials with a well-defined three-dimensional structure; example: PAMAM dendrimer.

Nanocomposites: Composite materials in which one or more phases are present in the nanoscale to improve material properties; examples: polymer–clay nanocomposite, CNT–polymer composite.

(Definition and one suitable example of each type – 1.75 × 4 = 7 marks.)

12. Compare the advantages of SEM and TEM.

- SEM provides three-dimensional surface morphology with a large depth of field, whereas TEM provides detailed internal structural information with much higher resolution.
- SEM can analyse bulk or thick specimens with relatively simple sample preparation, whereas TEM requires ultrathin specimens for electron transmission.
- SEM is mainly used for studying surface topography and morphology, whereas TEM is used for analysing crystal structure, lattice defects and nanoparticle size.

(Any 3 points of difference -7 marks.)

13. Write the working principle and procedure of chemical precipitation method for nanomaterial synthesis. How pH, temperature and precursor concentration influence particle size and morphology of synthesised nanomaterials?

Working principle: Nanoparticles are formed by supersaturation, nucleation and crystal growth when a precipitating agent is added to a precursor solution

Procedure: A precursor solution is prepared, followed by the addition of a precipitating agent under continuous stirring to induce precipitation; the precipitate is then aged, separated by filtration or centrifugation, washed, dried and, if required, calcined to obtain nanomaterials.

Effect of pH, temperature and precursor concentration: The pH, temperature and precursor concentration control the nucleation and crystal growth processes, thereby determining the size, shape and surface texture (morphology) of the synthesised nanoparticles

(principle -3 marks, procedure-3 marks, effect of pH, temperature and concentration -1 mark)

14. What is X-ray Photoelectron Spectroscopy. Explain its importance in nanotechnology.

Definition: X-ray Photoelectron Spectroscopy (XPS) is a surface analytical technique based on the photoelectric effect, in which X-rays eject electrons from a material to determine its surface composition.

Importance: XPS is used to determine the elemental composition, chemical/oxidation state and surface chemical bonding of nanomaterials.

Applications: It is widely used for surface characterization, thin-film analysis, nanomaterial quality control, and catalyst and semiconductor studies.

(Definition-3 marks, Importance/application-4 marks)

Part C
Essay Type Questions
Answer any 1 questions. Each question carries 15 marks

15. Explain Transmission Electron Microscopy discussing its principle, applications, advantages and basic instrumental features.

- **Principle:** TEM works on the principle that a beam of high-energy electrons is transmitted through an ultrathin specimen to produce a highly magnified image of its internal structure.
- **Basic instrumental features:** Electron gun, electromagnetic condenser lenses, specimen holder, objective and projector lenses, fluorescent screen/CCD camera, and vacuum system.
- **Advantages:** Very high magnification and resolution; provides detailed internal structure, crystal defects and lattice information.
- **Applications:** Used for nanoparticle size and morphology analysis, crystal structure determination, defect analysis, nanomaterials and biological specimen characterization.

(Principle-5 marks, instrumentation-5 marks, advantages-2.5 marks, application- 2.5 marks)

16. Explain the top down approaches used in nanochemistry. Discuss its methods, principle, advantages and limitations with suitable examples.

- **Definition & Principle:** Top-down approach is the fabrication of nanomaterials by reducing bulk materials into nanosized particles using physical or mechanical methods.
- **Methods:**
  - **Ball milling:** Bulk material is mechanically ground into nanoparticles using high-energy ball mills.
  - **Lithography:** Nanostructures are fabricated by selectively removing or patterning material using light or electron beams.

- **Laser ablation:** A high-energy laser beam vaporizes a solid target to produce nanoparticles.

• **Advantages:** Simple and scalable process; suitable for large-scale production; widely applicable to different materials.

• **Limitations:** Possibility of contamination and poor control over particle size and shape; defects may be introduced during processing.

• **Examples:** Nanoparticles prepared by ball milling, semiconductor nanostructures fabricated by photolithography/electron-beam lithography, and metal nanoparticles produced by laser ablation.

(Definition/principle-2 marks, Any two different methods-8 marks, advantages -2 marks, limitation-2 marks, example- 1 mark)